

BS – 101 Engineering Physics

Lecture – 3

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Flux of a Uniform Electric Field

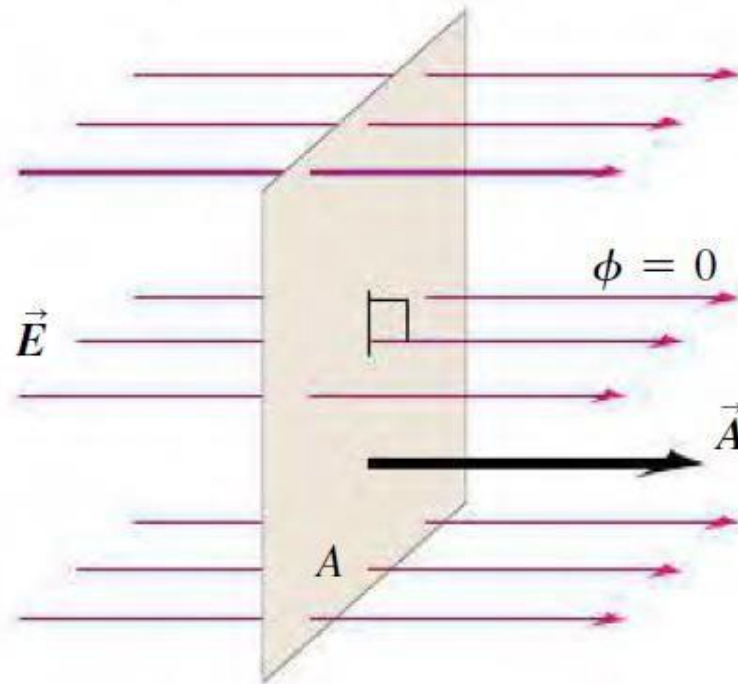
- Rate of flow of electric field through a given area.
- Flux per unit area.
- $E = \frac{\varphi_E}{A}$
- Unit:
 - Weber (W)
 - Nm^2/C
- $\varphi_E = EA$

Flux of a Uniform Electric Field (Case – 1)

Surface is face-on to electric field:

$\vec{E}$ and $\vec{A}$ are parallel (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 0$).

The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA$.

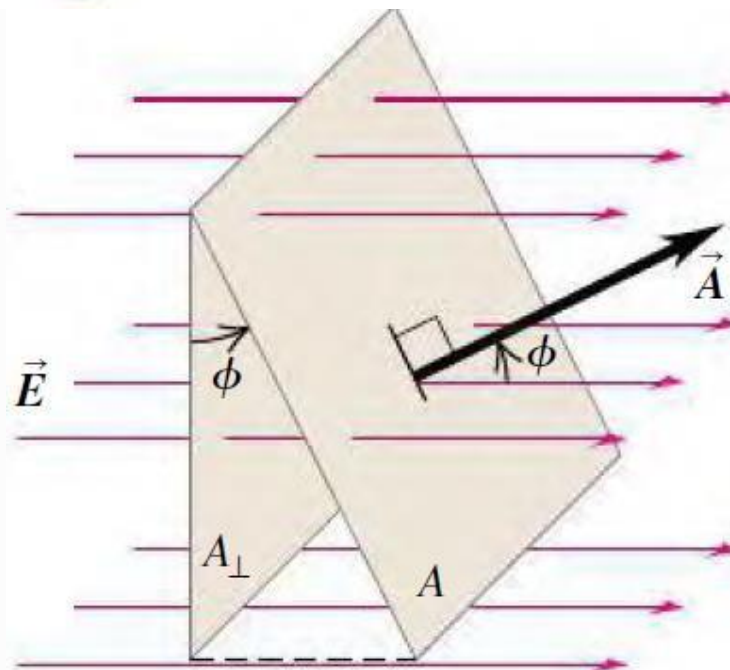


Flux of a Uniform Electric Field (Case – 2)

Surface is tilted from a face-on orientation by an angle ϕ :

The angle between $\vec{E}$ and $\vec{A}$ is ϕ .

The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \phi$.

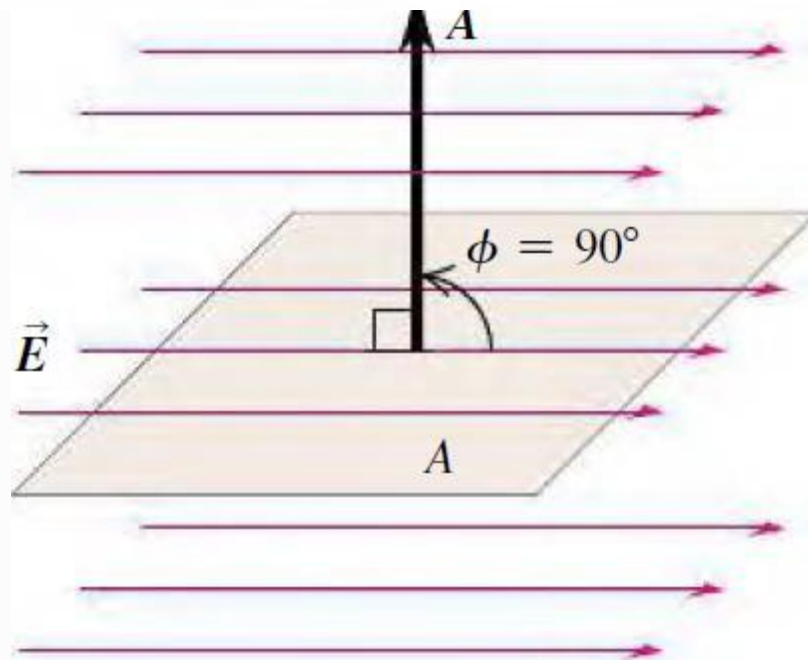


Flux of a Uniform Electric Field (Case – 3)

Surface is edge-on to electric field:

$\vec{E}$ and $\vec{A}$ are perpendicular (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 90^\circ$).

The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos 90^\circ = 0$.





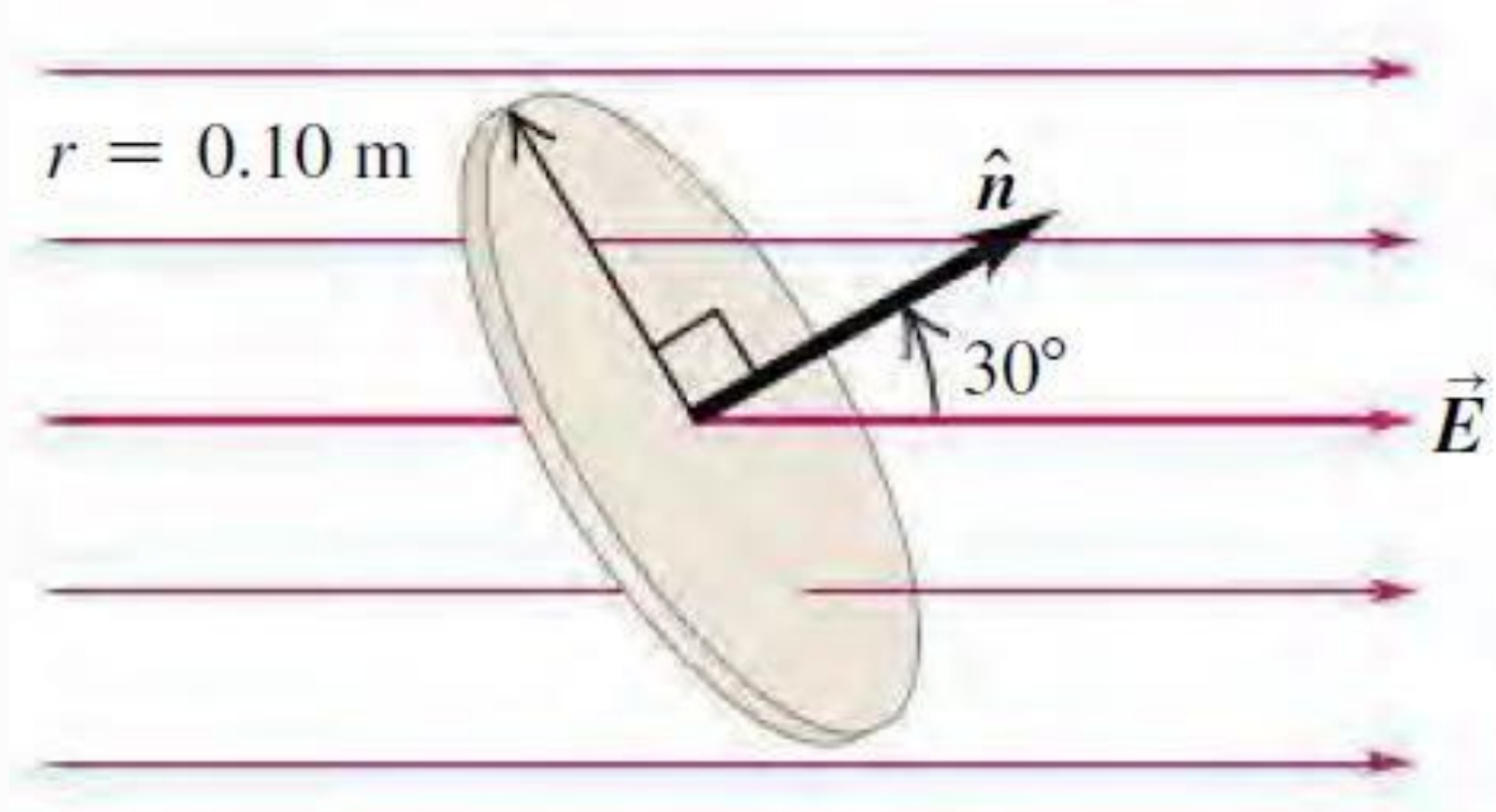
Direction of Area Vector

- The direction of a vector area $A \hat{n}$ could be represented by using a unit vector $\hat{n}$ perpendicular to the area. $\hat{n}$ is normal to the area.

Electric Flux through a Disk

- A disk of radius 0.10 m is oriented with its normal unit vector $\hat{n}$ at 30° to a uniform electric field $\vec{E}$ of magnitude 2.0×10^3 N/C. (Since this isn't a closed surface, it has no "inside" or "outside." That's why we have to specify the direction of $\hat{n}$ in the figure.) (a) What is the electric flux through the disk? (b) What is the flux through the disk if it is turned so that $\hat{n}$ is perpendicular to $\vec{E}$?
(c) What is the flux through the disk if $\hat{n}$ is parallel

Electric Flux through a Disk

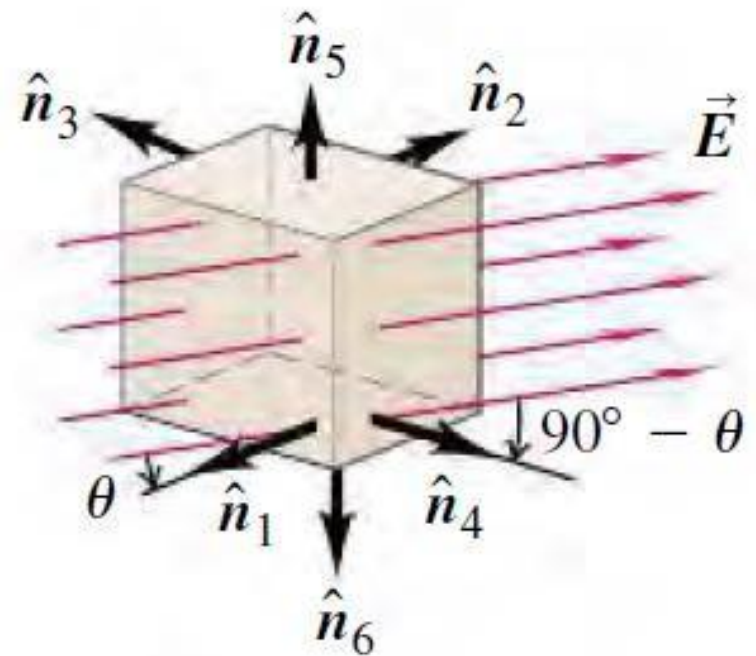
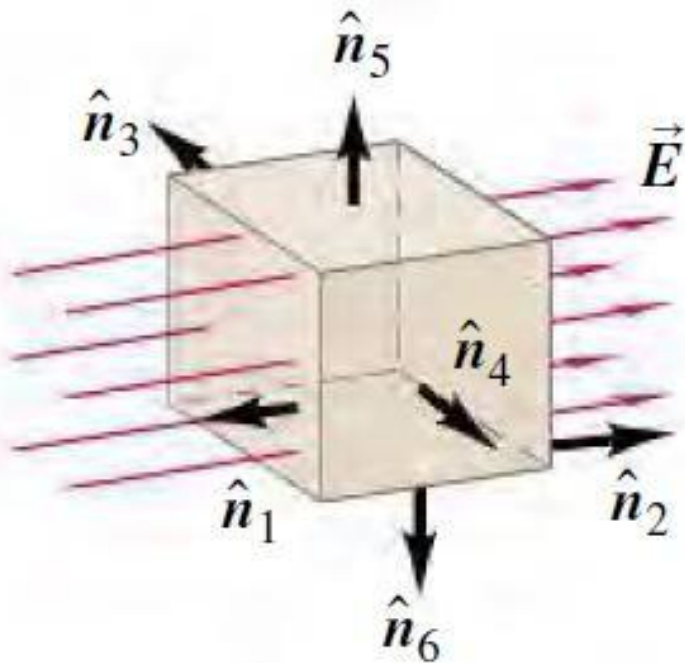




Electric Flux through a Cube

- An imaginary cubical surface of side L is in a region of uniform electric field $\vec{E}$. Find the electric flux through each face of the cube and the total flux through the cube when (a) it is oriented with two of its faces perpendicular to $\vec{E}$ and (b) the cube is turned by an angle θ about a horizontal axis.

Electric Flux through a Cube



Electric Flux through a Sphere

- A point charge $q = +3.0 \mu\text{C}$ is surrounded by an imaginary sphere of radius $r = 0.20 \text{ m}$ centered on the charge. Find the resulting electric flux through the sphere.

